Attempts were made to analyse the initial rate data using both integrated rate equations and the initial rate method, however the scatter of both the pseudo-first order analysis and the log rate vs. initial concentration plots were such that the results were statistically meaningless.

5.4 Conclusions

In this Chapter the solventless esterification of glycerol with the addition of ChCl has been studied for the first time. This work has shown that the addition of ChCl to glycerol can significantly affect the product distribution for the esterification of glycerol with lauric acid. Specific mono-esters and di-esters can be achieved by varying the liquid components. This site specific esterification is thought to occur due to the ability of ChCl to break up the hydrogen bonding in glycerol, but also to interact with water molecules produced in the reaction. The ChCl also decreases the viscosity of the liquid and at low acid concentrations when the system has not reached equilibrium, it is suggested that the process is mass transport controlled.

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CHAPTER 6

Conclusions and Future Work

6.1 Introduction

6.1.1 Physical Properties of Alcohol Based Deep Eutectic Solvents

6.1.2 Investigation of Intramolecular Interactions in Deep Eutectic Solvents Using Solvatochromic Probes

6.1.3 Esterification of Glycerol in a Choline Chloride Based Deep Eutectic Solvents

6.2 Future Work

6.1 Summary

This study has brought about a better understanding in the relatively novel area of alcohol based Deep Eutectic Solvents (DES). Several physical properties were investigated and the data was applied to several models to explain the behaviour of the systems.

6.1.1 Physical Properties of Alcohol Based Deep Eutectic Solvents

A series of polyol systems containing various concentrations of ChCl were investigate. The physical properties viscosity, conductivity, surface tension and density were investigated. For all diol systems, an increase in the ChCl concentration resulted in an increase in the viscosity of the system, whereas in the triol system, glycerol, the addition of ChCl caused a marked reduction. It was therefore shown that ChCl could be used to break up larger intermolecular structures, granting more degrees of freedom throughout the structure. This is also backed up by the density data which decreases as ChCl is added, where this is not true for the diol systems.

Conductivity data for the diol systems has shown that the point at which the system becomes a saturated salt solution is at 20 mol% ChCl, because at higher concentrations the conductivity decreases despite the increase in number of charge carrying species. This would suggest that the additional ChCl molecules are causing a hindrance to the flow of ions throughout the system

It was shown that it is possible to form eutectic mixtures from many different types of alcohols. A mono alcohol, phenol, was used to form an extremely fluid eutectic at a ratio of 4:1 phenol to ChCl. This eutectic offers some of the lowest viscosity data seen for any DES. This is most likely due to the delocalised π-electron system on the benzene ring which is increasing the amount of charge delocalisation on ChCl. On the other hand, polyols, with more than 3 OH moieties were also investigated and were shown to form DES's with ChCl. Sugars such as erythritol, xylitol, D-(-) fructose and D-glucose all formed stable DES's which opens up the potential for creating benign and possibly edible liquids from naturally occurring salts and sugars.

6.1.2 Investigation of Intramolecular Interactions in Deep Eutectic Solvents Using Solvatochromic Probes

In this work several parameters, E_T^N, α, β and π* have been obtained for the first time for several alcohol based DESs. The data has given information on the relative polarity of these systems, which suggests that they are similar to $RNH_3^+X^-$, $R_2NH_2^+X^-$ and imidazolium ionic liquids. It was also shown that the effect of ChCl on these systems is very much dependent upon the nature of the hydrogen bond donor. Whilst the addition of ChCl gave an increase in the E_T^N value for every system, the degree to which it was altered was not the same in each case. The greatest changes were seen in the 1,3- and 1,2-butanediol systems.

The addition of ChCl has slightly different effects on the values for the hydrogen bond acceptor paramerter β. In the diol cases, with the exception of 1,4-butanediol, the β value decreases when ChCl is added, which would be expected since the Cl^- will be interacting with the OH moieties, reducing the ability to further accept hydrogen bonds. In the glycerol systems, there is no change. This would perhaps indicate that the number of free OH groups for the indicator probe to interact with does not change when ChCl is added, despite its addition increasing the degree of freedom throughout the system. This could indicate that one OH moiety is always free in both glycerol and the eutectic mixture. The addition of ChCl shows an increase in the values for hydrogen bond donor α.

It has been proposed that there is a correlation between the values for $E_T(30)$, α and β with regards to the molar volume per OH moiety. There are distinct trends, distinguishing between the pure alcohol systems and the eutectic systems. This is not surprising since the OH moiety is a hydrogen bond donor/acceptor. There was no such trend observed for the π* data.

6.1.3 Esterification of Glycerol in a Choline Chloride Based Deep Eutectic Solvents

Solventless esterification of glycerol and lauric was carried out using ChCl with the intention of using its hydrogen bonding interaction with glycerol to behave a

potential site inhibitor as to selectively form mono- or di-ester products. The work has shown that the addition of ChCl to the system actually encourages the reaction to progress further towards the di-ester product, as opposed to the pure glycerol system encouraging only mono-ester formation. It is thought that because ChCl breaks up the 3-D structure of glycerol, this actually works in favour of the esterification reaction as it can further react with glycerol molecules already esterified. It is also possible that ChCl is interacting with water molecules produced in the reaction, effectively removing them from the equilibrium, which in turn helps drive the reaction forward. The ChCl also decreases the viscosity of the liquid and at low acid concentrations when the system has not reached equilibrium, it is suggested that the process is mass transport controlled.

6.2 Future Work

There is still a great deal that could be learnt from these systems with some further work. In a continuation of the physical properties of the alcohol based DES's, additives could be added to the system and the changes in these properties could be investigated. The addition of water would be interesting to investigate because the hygroscopic nature of ChCl means that these systems are constantly absorbing moisture from the atmosphere over time. Mixed systems containing alcohols with dicarboxylic acids/diamides could also be investigated.

The technique EXAFS (Extended X-Ray Absorption Fine Structure) could be employed to gather information on the interaction between the Cl and OH moieties and see how this interaction changes in each of the systems investigated in this thesis. The technique has been used to help understand the local structure in many systems such as glass, amorphous and liquid systems, organometallic compounds, metal clusters. This interaction between the chloride anion and hydroxyl group is crucial in the formation of DES's but has not been fully investigated due to the inherent difficulties of being able to do so.

A more sensitive method for investigating the phase transitions of the systems described throughout this thesis would be to use a DSC (Differential Scanning Calorimtery), which would give a more accurate account of the melting and glass

transitions, for example. The DSC measures energy or heat capacity changes and so would give further qualitative data on these systems.

While it would be expected that the Lewis basicity would change with the addition of ChCl to the alcohol systems, none of the parameters determined in Chapter 4 highlight this. It is proposed that a new scale of chloride activity could be developed by using an indicator that is sensitive to Cl^- ions. Some transition metal hydrate salts are well known to change colour when heated or in the presence of ligands stronger than H_2O because they lose waters of hydration from their coordination sphere. Examples of this include $CuSO_4.5H_2O$ (blue to yellow) or $CoCl_2.6H_2O$ (blue to pink). In an endeavour to develop a chloride activity scale for chloride eutectics $CuSO_4.5H_2O$ could be added to the liquids and the ratio of the two absorbance bands ($[Cu.4H_2O]^{2+}$ and $CuCl_3^-$) could be measured by UV-Vis spectroscopy.

Appendix

7.1 Physical Properties Raw Data

7.1.1 Choline Chloride : 1,2-Ethanediol

7.1.2 Choline Chloride : 1,4-Butanediol

7.1.3 Choline Chloride : 1,6-Hexanediol

7.1.4 Choline Chloride : 1,2 Butanediol

7.1.5 Choline Cholride : 1,3 Butanediol

7.1.6 Choline Chloride : 2,3 Butanediol

7.1.7 Choline Chloride : Glycerol

7.2 Esterification Raw Data

7.2.1 Pure Glycerol

7.2.2 ChCl : Glycerol 1:3

7.2.3 ChCl : Glycerol 1:2

7.2.4 ChCl : Glycerol 1:1.5

7.2.5 ChCl : Glycerol 1:1

7.3 Publications

7.1 Physical Properties Raw Data

7.1.1 Choline Chloride : 1,2-Ethanediol

Table 7.1: *Raw data for 1,2-ethanediol with 5% choline chloride*

Temperature (^oC)	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**	**20**
Viscosity (cP)	43	35	30	26	23	21	19	17	15
Conductivity (mS)	2.59	2.68	2.82	2.99	3.14	3.29	3.45	3.71	3.93
Surface Tension (n/nM)	48.0	47.8	47.4	47.1	46.8	46.4	46.2	45.8	45.7

Table 7.2: *Raw data for 1,2-ethanediol with 10% choline chloride*

Temperature (^oC)	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**	**20**
Viscosity (cP)	45	37	32	28	25	23	21	19	17
Conductivity (mS)	4.31	4.43	4.57	4.72	4.89	5.09	5.33	5.58	5.83
Surface Tension (n/nM)	49.5	49.3	48.9	48.6	48.3	48.0	47.8	47.6	47.5

Table 7.3: *Raw data for 1,2-ethanediol with 15% choline chloride*

Temperature (^oC)	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**	**20**
Viscosity (cP)	48	40	35	31	28	25	23	21	19
Conductivity (mS)	5.27	5.37	5.52	5.71	5.96	6.28	6.62	6.98	7.35
Surface Tension (n/nM)	51.4	51.2	51.0	50.7	50.4	50.2	49.9	49.7	49.6

Table 7.4: *Raw data for 1,2-ethanediol with 20% choline chloride*

Temperature (^oC)	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**	**20**
Viscosity (cP)	53	46	40	36	31	28	25	23	21
Conductivity (mS)	5.68	5.79	5.93	6.11	6.35	6.59	6.84	7.13	7.56
Surface Tension (n/nM)	50.6	50.2	50.0	49.8	49.7	49.4	49.3	49.2	49.0

Table 7.5: *Raw data for 1,2-ethanediol with 25% choline chloride*

Temperature (^oC)	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**	**20**
Viscosity (cP)	68	60	54	49	44	40	36	32	29
Conductivity (mS)	5.40	5.51	5.63	5.77	5.98	6.24	6.61	6.97	7.44
Surface Tension (n/nM)	50.8	50.4	50.0	49.6	49.4	49.2	48.9	48.6	48.3

Table 7.6: *Raw data for 1,2-ethanediol with 30% choline chloride*

Temperature (^oC)	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**	**20**
Viscosity (cP)	230	171	133	101	78	58	48	42	37
Conductivity (mS)	5.05	5.12	5.23	5.35	5.50	5.74	6.05	6.40	6.89
Surface Tension (n/nM)	49.6	49.2	48.9	48.7	48.4	48.2	48.0	47.7	47.5

Table 7.7: *Raw data for 1,2-ethanediol with 33% choline chloride*

Temperature (^oC)	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**	**20**
Viscosity (cP)	266	197	144	115	91	67	50	43	38
Conductivity (mS)	4.90	5.02	5.15	5.30	5.48	5.67	5.93	6.16	6.58
Surface Tension (n/nM)	50.0	49.8	49.4	49.2	49.1	48.6	48.3	47.9	47.7

Table 7.8: *Raw data for density of the 1,2-ethanediol systems choline chloride at 293K*

ChCl %	0	5	10	15	20	25	30	33
Density (gcm^{-3})	1.113	1.114	1.115	1.115	1.117	1.118	1.118	1.120

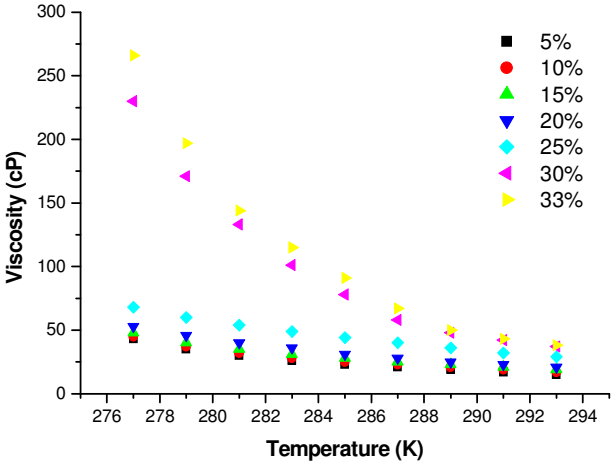


Figure 7.1*: Plot of temperature versus viscosity as a function of choline chloride composition for the 1,2-ethanediol system*

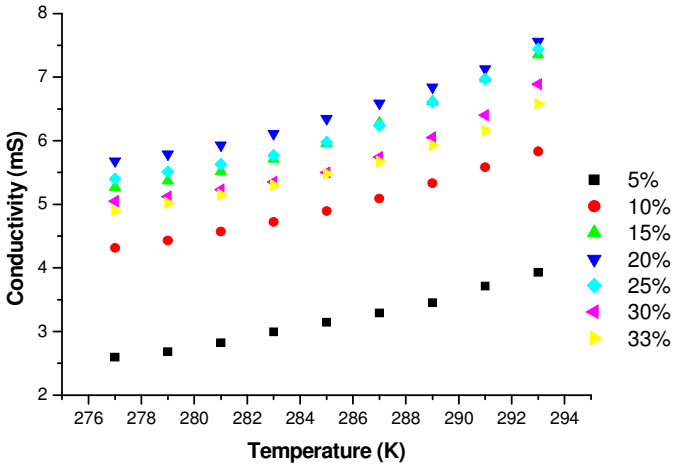


Figure 7.2*: Plot of temperature versus conductivity as a function of choline chloride composition for the 1,2-ethanediol system*

7.1.2 Choline Chloride : 1,4-Butanediol

Table 7.9: *Raw data for 1,4-butanediol with 5% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	78	68	60	54	48	44	40	36	32	29
Conductivity (µS)	593	617	640	671	708	746	784	824	865	910
Surface Tension (n/nM)	46.4	46.3	46.1	46.0	45.9	45.8	45.7	45.6	45.4	45.3

Table 7.10: *Raw data for 1,4-butanediol with 10% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	84	72	60	54	48	44	40	36	33	30
Conductivity (µS)	923	948	974	1009	1068	1144	1209	1308	1422	1563
Surface Tension (n/nM)	46.8	46.7	46.6	46.4	46.3	46.3	46.2	46.1	46.0	45.8

Table 7.11: *Raw data for 1,4-butanediol with 15% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	78	68	62	55	49	44	40	37	34	31
Conductivity (µS)	1271	1323	1378	1443	1529	1610	1719	1845	1990	2170
Surface Tension (n/nM)	46.9	46.8	46.8	46.7	46.5	46.4	46.3	46.1	46.0	46.0

Table 7.12: *Raw data for 1,4-butanediol with 20% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	88	76	68	61	55	50	46	42	38	35
Conductivity (µS)	1606	1664	1739	1829	1947	2080	2230	2360	2500	2710
Surface Tension (n/nM)	47.4	47.3	47.1	47.0	46.9	46.8	46.6	46.6	46.3	46.2

Table 7.13: *Raw data for 1,4-butanediol with 25% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	176	136	98	76	66	58	52	47	42	38
Conductivity (µS)	1654	1683	1748	1865	1998	2130	2280	2410	2630	2880
Surface Tension (n/nM)	47.6	47.5	47.4	47.4	47.3	47.3	47.2	47.1	47.1	47.0

Table 7.14: *Raw data for density of the 1,4-butanediol systems choline chloride at 293K*

ChCl %	**0**	**5**	**10**	**15**	**20**	**25**
Density (gcm^{-3})	1.017	1.021	1.026	1.036	1.046	1.052

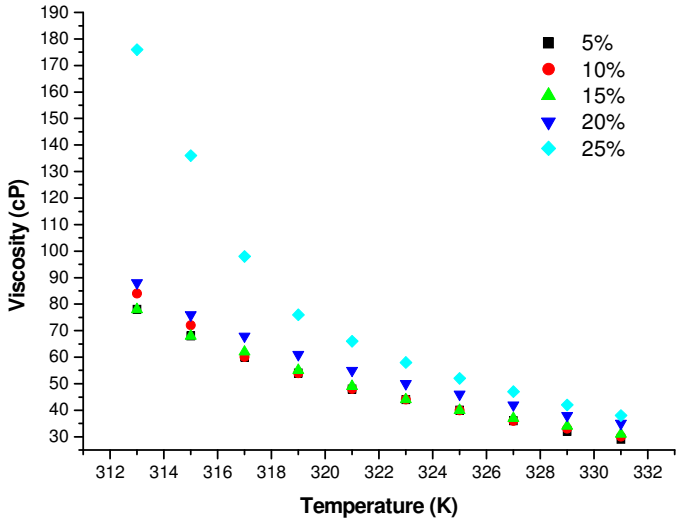


***Figure 7.3**: Plot of temperature versus viscosity as a function of choline chloride composition for the 1,4-butanediol system*

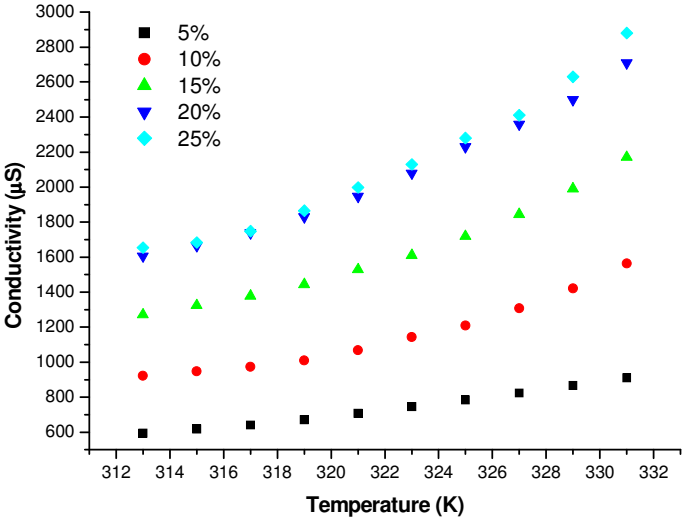


***Figure 7.4**: Plot of temperature versus conductivity as a function of choline chloride composition for the 1,4-butanediol system*

7.1.3 Choline Chloride : 1,6-Hexanediol

Table 7.15*: Raw data for phase diagram plot of choline chloride : 1,6-hexanediol*

Molar Ratio (1,6-Hexanediol: Choline Chloride)	Freezing Point (^{o}C)
1:1	165
2:1	72
3:1	31
4:1	27
5:1	28
6:1	29
7:1	30
8:1	31

Table 7.16: *Raw data for 1,6-hexanediol with 5% choline chloride*

** note surface tension is measured at a different temperature than viscosity and conductivity*

Temperature (^{o}C)	**45**	**47.5**	**50**	**52.5**	**55**	**57.5**	**60**	**62.5**
Viscosity (cP)	43	36	33	29	26	24	22	20
Conductivity (µS)	317	333	349	370	393	417	440	466
Temperature (^{o}C)	**43.5**	**46**	**48.5**	**51**	**53.5**	**56**	**58.5**	**61.5**
Surface Tension (n/nM)	42.3	42.1	41.9	41.5	41.3	41.2	41.1	41.0

Table 7.17: *Raw data for 1,6-hexanediol with 10% choline chloride*

** note surface tension is measured at a different temperature than viscosity and conductivity*

Temperature (^{o}C)	**45**	**47.5**	**50**	**52.5**	**55**	**57.5**	**60**	**62.5**
Viscosity (cP)	43	38	34	30	27	24	22	20
Conductivity (µS)	549	576	607	639	680	713	759	808
Temperature (^{o}C)	**43.5**	**46**	**48.5**	**51**	**53.5**	**56**	**58.5**	**61.5**
Surface Tension (n/nM)	42.6	42.4	42.3	41.8	41.7	41.6	41.5	41.3

Table 7.18: *Raw data for 1,6-hexanediol with 15% choline chloride*

** note surface tension is measured at a different temperature than viscosity and conductivity*

Temperature (^oC)	**45**	**47.5**	**50**	**52.5**	**55**	**57.5**	**60**	**62.5**
Viscosity (cP)	43	38	34	31	28	25	22	20
Conductivity (µS)	867	882	929	978	1035	1095	1187	1304
Temperature (^oC)	**43.5**	**46**	**48.5**	**51**	**53.5**	**56**	**58.5**	**61.5**
Surface Tension (n/nM)	42.8	42.6	42.5	42.4	42.2	42.2	42.0	42.0

Table 7.19: *Raw data for 1,6-hexanediol with 20% choline chloride*

** note surface tension is measured at a different temperature than viscosity and conductivity*

Temperature (^oC)	**45**	**47.5**	**50**	**52.5**	**55**	**57.5**	**60**	**62.5**
Viscosity (cP)	48	43	38	34	31	29	26	23
Conductivity (µS)	1090	1126	1186	1250	1332	1446	1565	1699
Temperature (^oC)	**43.5**	**46**	**48.5**	**51**	**53.5**	**56**	**58.5**	**61.5**
Surface Tension (n/nM)	43.2	43.1	43.0	42.9	42.6	42.6	42.5	42.3

Table 7.20: *Raw data for 1,6-hexanediol with 25% choline chloride*

** note surface tension is measured at a different temperature than viscosity and conductivity*

Temperature (^oC)	**45**	**47.5**	**50**	**52.5**	**55**	**57.5**	**60**	**62.5**
Viscosity (cP)	48	43	38	34	31	28	25	22
Conductivity (µS)	1325	1367	1428	1497	1589	1693	1823	1968
Temperature (^oC)	**43.5**	**46**	**48.5**	**51**	**53.5**	**56**	**58.5**	**61.5**
Surface Tension (n/nM)	43.6	43.5	43.4	43.3	43.1	42.9	42.7	42.7

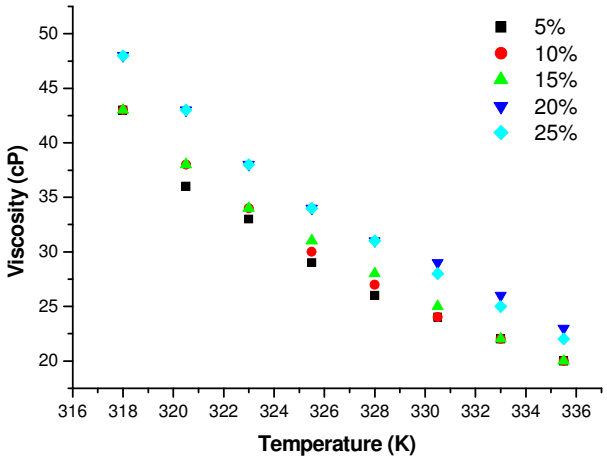


***Figure 7.5**: Plot of temperature versus viscosity as a function of choline chloride composition for the 1,6-hexanediol system*

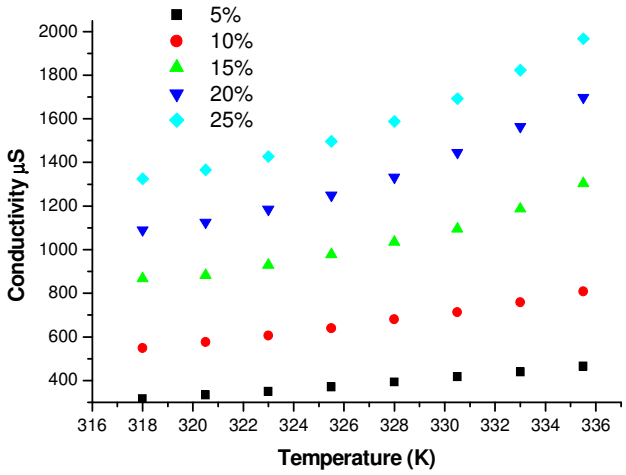


***Figure 7.6**: Plot of temperature versus viscosity as a function of choline chloride composition for the 1,6-hexanediol system*

7.1.4 Choline Chloride : 1,2 Butanediol

Table 7.21: *Raw data for 1,2-butanediol with 5% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	62	55	48	41	34	31	26	22	19	17
Conductivity (µS)	459	489	524	564	606	645	685	730	775	820
Surface Tension (n/nM)	32.9	32.8	32.5	32.3	32.1	31.9	31.7	31.6	31.3	31.1

Table 7.22: Raw data for 1,2-butanediol with 10% choline chloride

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	67	53	45	41	36	31	26	22	19	17
Conductivity (µS)	811	877	937	1010	1071	1143	1226	1291	1364	1441
Surface Tension (n/nM)	33.7	33.4	33.2	33.0	32.8	32.6	32.2	32.0	31.9	31.7

Table 7.23: *Raw data for 1,2-butanediol with 15% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	70	60	53	48	38	36	29	26	22	19
Conductivity (µS)	1078	1167	1245	1309	1415	1508	1674	1770	1870	1988
Surface Tension (n/nM)	34.2	33.9	33.7	33.4	33.3	33.2	33.1	33.0	32.9	32.6

Table 7.24: *Raw data for 1,2-butanediol with 20% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	79	70	62	55	48	41	36	31	29	26
Conductivity (µS)	1358	1429	1524	1623	1716	1920	2030	2150	2260	2380
Surface Tension (n/nM)	34.7	34.5	34.3	34.2	34.1	33.9	33.7	33.5	33.4	33.2

Table 7.25: *Raw data for density of the 1,2 -butanediol systems choline chloride at 293K*

ChCl %	**0**	**5**	**10**	**15**	**20**
Density (gcm^{-3})	1.006	1.009	1.013	1.022	1.030

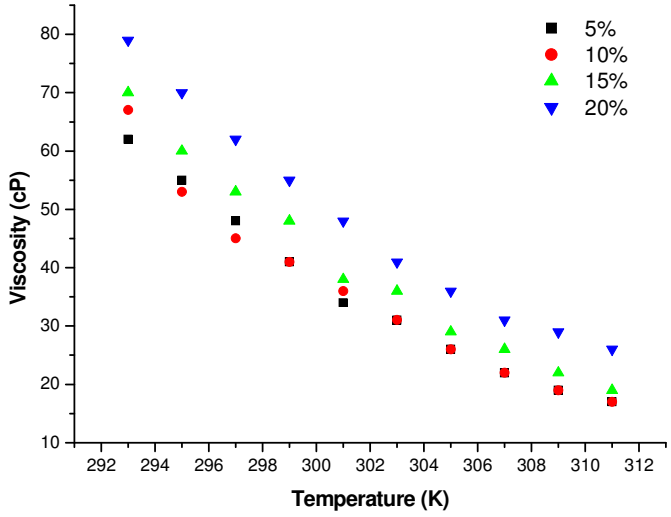


Figure 7.7*: Plot of temperature versus viscosity as a function of choline chloride composition for the 1,2-butanediol system*

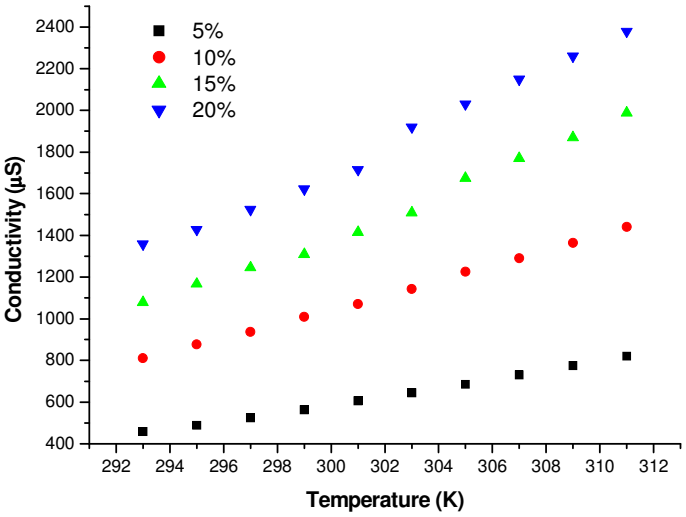


***Figure 7.8**: Plot of temperature versus conductivity as a function of choline chloride composition for the 1,2-butanediol system*

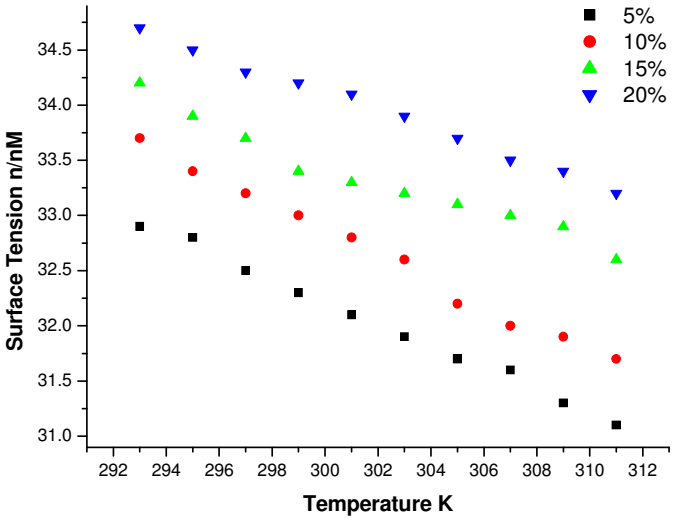


***Figure 7.9**: Plot of temperature versus surface tension as a function of choline chloride composition for the 1,2-butanediol system*

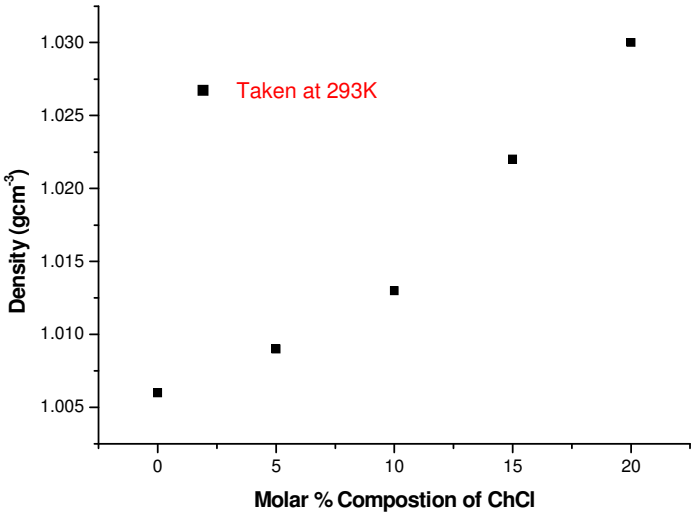


***Figure 7.10**: Plot of choline chloride composition versus density for the 1,2-butanediol system taken at 293 K*

7.1.5 Choline Cholride : 1,3 Butanediol

***Table 7.26:** Raw data for 1,3-butanediol with 5% choline chloride*

Temperature (°C)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	98	86	77	67	60	53	46	41	36	31
Conductivity (µS)	351	375	403	431	460	489	521	554	587	622
Surface Tension (n/nM)	34.4	34.2	34.0	33.9	33.7	33.4	33.0	32.6	32.3	31.9

***Table 7.27:** Raw data for 1,3-butanediol with 10% choline chloride*

Temperature (°C)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	101	89	79	70	62	55	48	43	38	34
Conductivity (µS)	645	693	741	792	846	900	953	1014	1069	1120
Surface Tension (n/nM)	38.2	37.9	37.8	37.5	37.5	36.9	36.5	36.2	35.9	35.7

Table 7.28: *Raw data for 1,3-butanediol with 15% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	106	94	82	72	62	53	48	43	36	31
Conductivity (µS)	934	987	1051	1118	1190	1264	1335	1406	1480	1616
Surface Tension (n/nM)	39.0	38.9	38.6	38.3	38.3	38.2	38.0	37.6	37.2	36.9

Table 7.29: *Raw data for 1,3-butanediol with 20% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	113	101	89	77	70	60	53	48	43	39
Conductivity (µS)	1152	1228	1309	1380	1475	1565	1652	1824	1923	2070
Surface Tension (n/nM)	40.1	39.8	39.6	39.6	39.2	39.2	38.9	38.5	38.3	38.1

Table 7.30: *Raw data for density of the 1,3 -butanediol systems choline chloride at 293K*

ChCl %	**0**	**5**	**10**	**15**	**20**
Density (gcm^{-3})	1.005	1.008	1.015	1.026	1.034

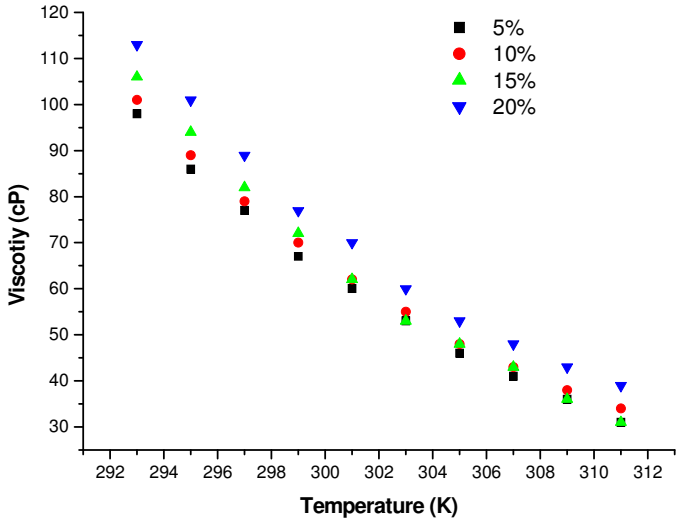


***Figure 7.11**: Plot of temperature versus viscosity as a function of choline chloride composition for the 1,3-butanediol system*

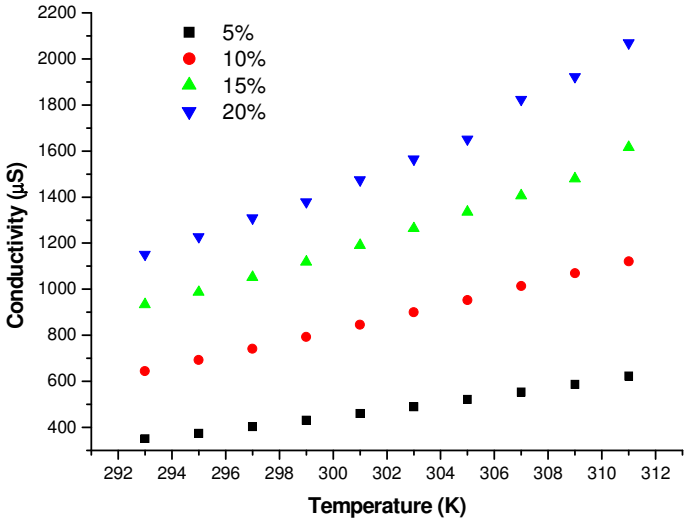


***Figure 7.12**: Plot of temperature versus conductivity as a function of choline chloride composition for the 1,3-butanediol system*

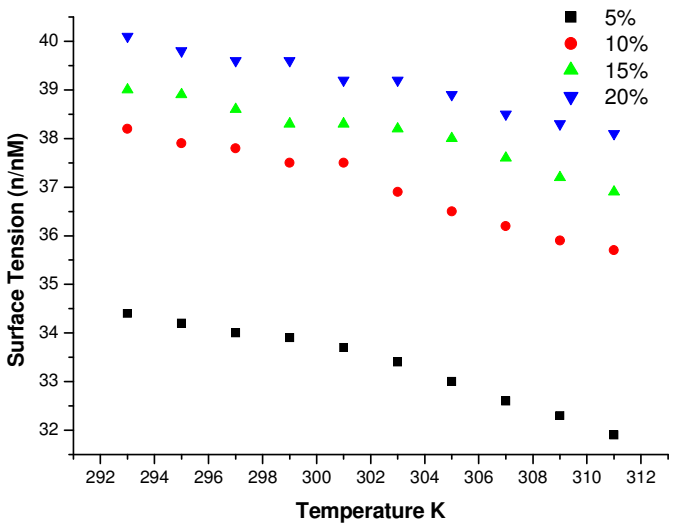


***Figure 7.13**: Plot of temperature versus surface tension as a function of choline chloride composition for the 1,3-butanediol system*

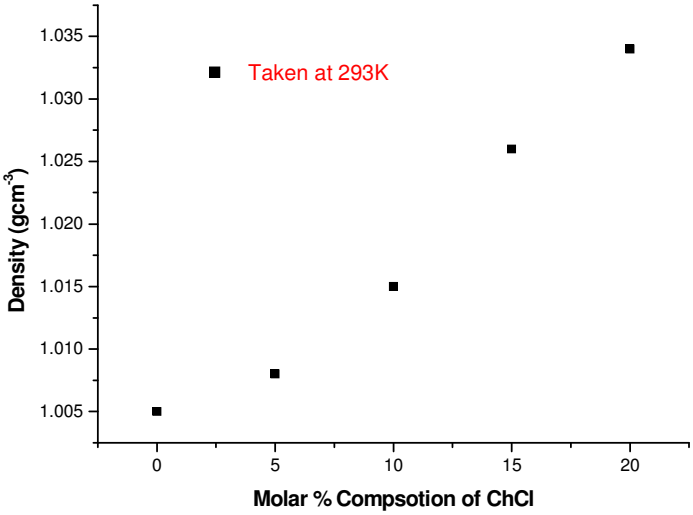


***Figure 7.14**: Plot of choline chloride composition versus density for the 1,3-butanediol system taken at 293 K*

7.1.6 Choline Chloride : 2,3 Butanediol

Table 7.31: *Raw data for 2,3-butanediol with 5% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	120	96	82	69	58	48	41	36	31	26
Conductivity (µS)	246	277	302	328	356	389	422	457	491	526
Surface Tension (n/nM)	34.2	34.0	33.7	33.7	33.5	33.2	33.0	32.9	32.6	32.3

Table 7.32: *Raw data for 2,3-butanediol with 10% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	122	101	86	72	62	53	46	38	34	31
Conductivity (µS)	476	530	575	623	672	734	788	840	898	958
Surface Tension (n/nM)	34.5	34.3	34.1	33.9	33.4	33.3	33.2	33.1	32.8	32.8

Table 7.33: *Raw data for 2,3-butanediol with 15% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	130	113	96	82	70	60	53	46	41	36
Conductivity (µS)	776	839	916	996	1076	1163	1237	1321	1403	1495
Surface Tension (n/nM)	35.1	34.9	34.6	34.4	34.2	34.0	33.7	33.5	33.3	33.0

Table 7.34: *Raw data for 2,3-butanediol with 20% choline chloride*

Temperature (^oC)	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**
Viscosity (cP)	178	132	115	98	86	74	65	55	48	43
Conductivity (µS)	895	956	1026	1108	1190	1281	1372	1467	1566	1751
Surface Tension (n/nM)	35.6	35.4	34.9	34.7	34.7	34.5	34.3	34.0	33.7	33.5

Table 7.35: *Raw data for density of the 2,3 -butanediol systems choline chloride at 293K*

ChCl %	**0**	**5**	**10**	**15**	**20**
Density (gcm^{-3})	0.995	1.001	1.015	1.024	1.032

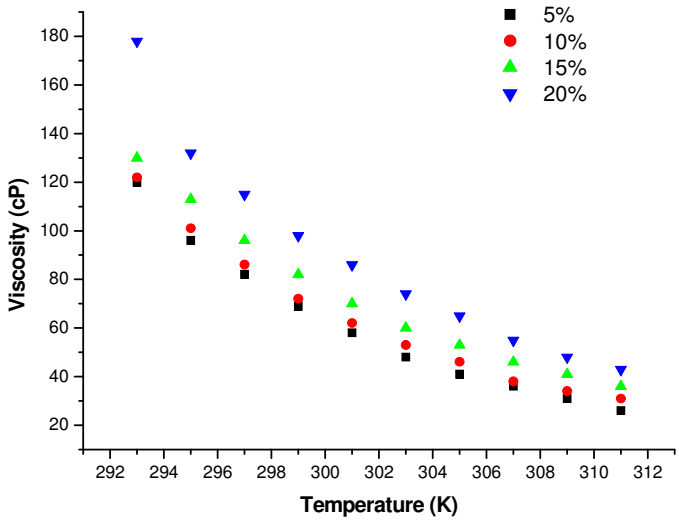


Figure 7.15*: Plot of temperature versus viscosity as a function of choline chloride composition for the 2,3-butanediol system*

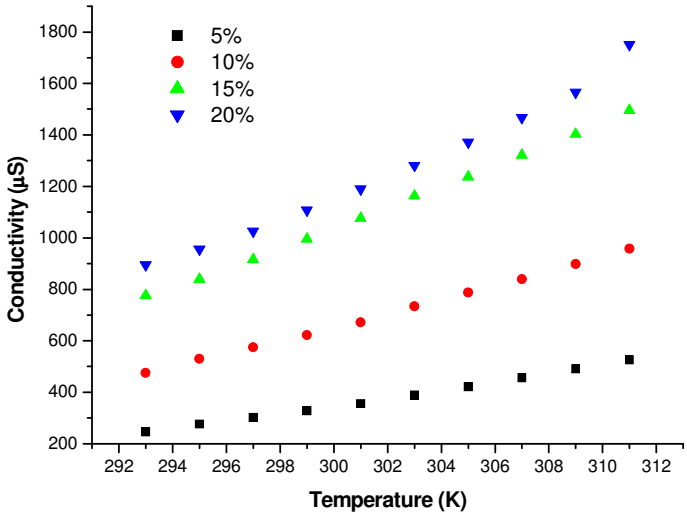


***Figure 7.16**: Plot of temperature versus conductivity as a function of choline chloride composition for the 2,3-butanediol system*

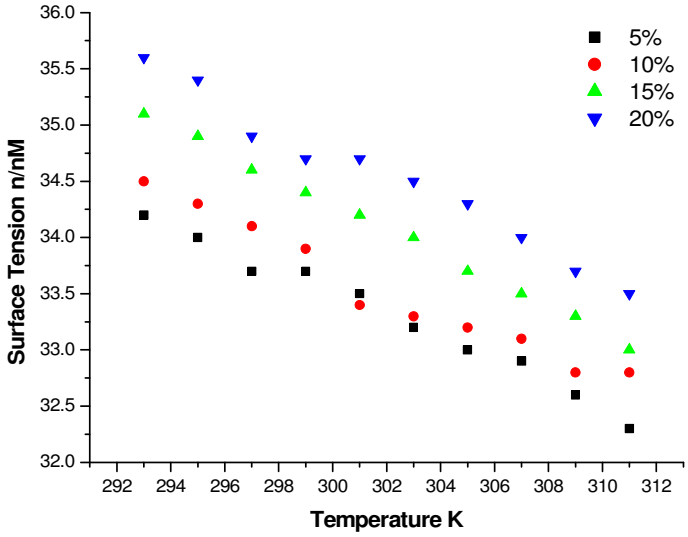


***Figure 7.17**: Plot of temperature versus surface tension as a function of choline chloride composition for the 2,3-butanediol system*

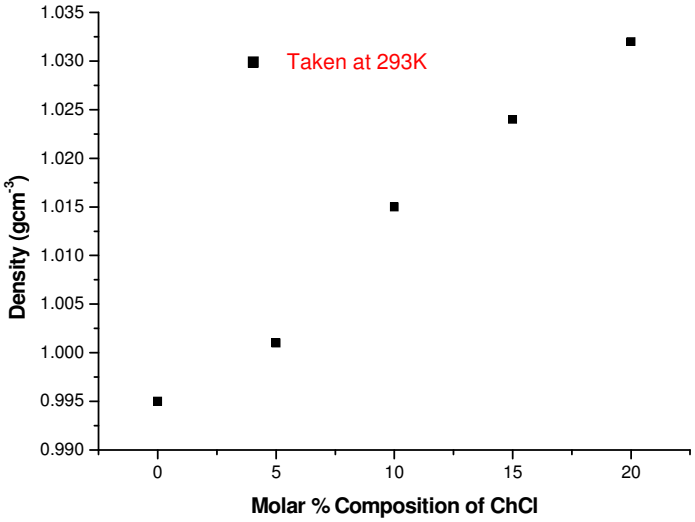


***Figure 7.18**: Plot of choline chloride composition versus density for the 2,3-butanediol system*

7.1.7 Choline Chloride : Glycerol

***Table 7.36:** Raw data for glycerol with 5% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	998	654	450	314	222	163	120	90
Conductivity (µS)	106.4	143.3	190.1	252	320	394	479	568
Surface Tension (n/nM)	63.7	62.8	62.0	61.3	60.5	59.8	59.3	58.7

***Table 7.37:** Raw data for glycerol with 10% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	790	503	349	247	180	132	98	74
Conductivity (µS)	243	297	374	483	601	759	980	1245
Surface Tension (n/nM)	61.6	61.0	60.3	59.5	59.0	58.4	57.9	57.5

Table 7.38: *Raw data for glycerol with 15% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	548	380	270	194	140	107	80	60
Conductivity (µS)	470	555	666	798	963	1114	1403	1788
Surface Tension (n/nM)	60.8	60.2	59.7	59.1	58.7	58.1	57.6	57.3

Table 7.39: *Raw data for glycerol with 20% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	503	350	246	178	132	98	86	57
Conductivity (µS)	580	739	930	1140	1362	1609	1923	2250
Surface Tension (n/nM)	57.4	57.0	56.6	56.1	55.7	55.2	54.8	54.4

Table 7.40: *Raw data for glycerol with 25% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	450	320	229	169	126	95	73	58
Conductivity (µS)	850	1030	1244	1490	1835	2140	2480	2930
Surface Tension (n/nM)	50.8	50.3	49.9	49.3	48.9	48.4	48.0	47.6